



Do different types of salt affect the environment?

Protocol

Materials required:

- 5 clear 3oz plastic cups with lids
- Sharpie marker
- Blue transfer loops
- Dish of *Lemna* plants
- Bottle of *Lemna* media solution
- Bottle of road salt solution
- Bottle of Sodium chloride (NaCl) solution
- Bottle of Potassium chloride (KCl) solution
- Bottle of Calcium chloride (CaCl₂) solution
- One 250mL graduated cylinder
- Four 25mL graduated cylinders

Procedure: Set up

1. Grab your five plastic cups and label all of them with your group name.
2. Also label each cup with the treatment you'll be measuring from that cup.
The five treatments are:
 - Positive control (+)
 - Negative control (-)
 - Sodium chloride (NaCl)
 - Potassium chloride (KCl)
 - Calcium chloride (CaCl₂)
3. Using the labeled 25mL graduated cylinders add 20mL of each of the four salt solutions to their graduate cylinders and respective cups.
4. Using the 250mL graduated cylinder provided, add 180mL of *Lemna* media to each of your four salt solution cups. This will make the total volume of these four cups 200mL.
5. Using the same 250mL graduated cylinder, add 200mL of *Lemna* media into your positive control cup.
6. Using the provided blue transfer loops, transfer 4 *Lemna* plants into each of your five cups. IMPORTANT make sure to add plants that only contain 3 leaves.
7. Once the correct number of plants have been added to all cups, place the lids on top to close the cups. Place the cups under the grow light in the lab; they will grow until your next visit!